Lagrange Interpolating Polynomials

Example 04-02a: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728

Lagrange Interpolating Polynomials

<u>Example 04-02a</u>: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728

$$p_1(x) = L_0^{(1)}(x) \cdot f(x_0) + L_1^{(1)}(x) \cdot f(x_1)$$

Lagrange Interpolating Polynomials

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$$p_1(x) = L_0^{(1)}(x) \cdot f(x_0) + L_1^{(1)}(x) \cdot f(x_1)$$

$$= \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1)$$

Lagrange Interpolating Polynomials

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$$= \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1)$$

$$f(0.45) \approx p_1(0.45) = \frac{0.45 - 0.7}{0.0 - 0.7} \cdot 0.0000000 + \frac{0.45 - 0.0}{0.7 - 0.0} \cdot 1.2039728$$

Lagrange Interpolating Polynomials

Example 04-02a: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

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$$f(0.45) \approx p_1(0.45) = \frac{0.45 - 0.7}{0.0 - 0.7} \cdot 0.0000000 + \frac{0.45 - 0.0}{0.7 - 0.0} \cdot 1.2039728$$

$$= 0.7739825$$

Lagrange Interpolating Polynomials

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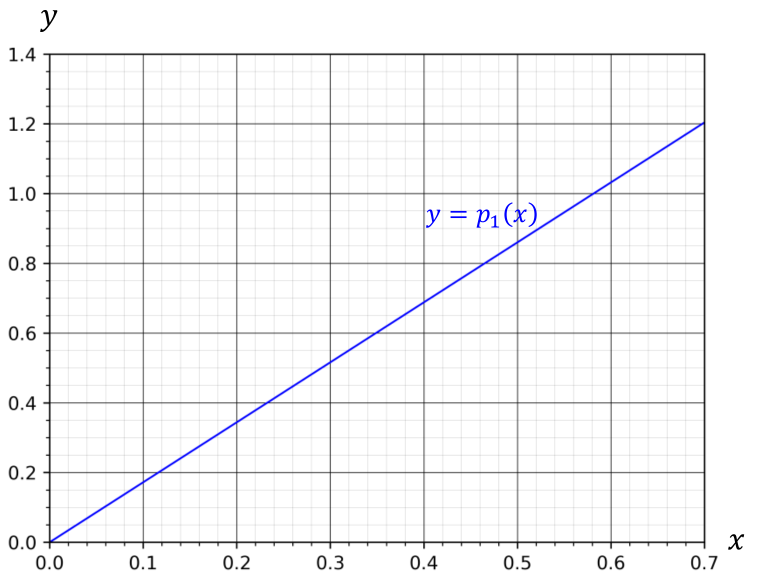
$$= \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1)$$

$$f(0.45) \approx p_1(0.45) = \frac{0.45 - 0.7}{0.0 - 0.7} \cdot 0.0000000 + \frac{0.45 - 0.0}{0.7 - 0.0} \cdot 1.2039728$$

$$= 0.7739825$$



Lagrange Interpolating Polynomials

Example 04-02a: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

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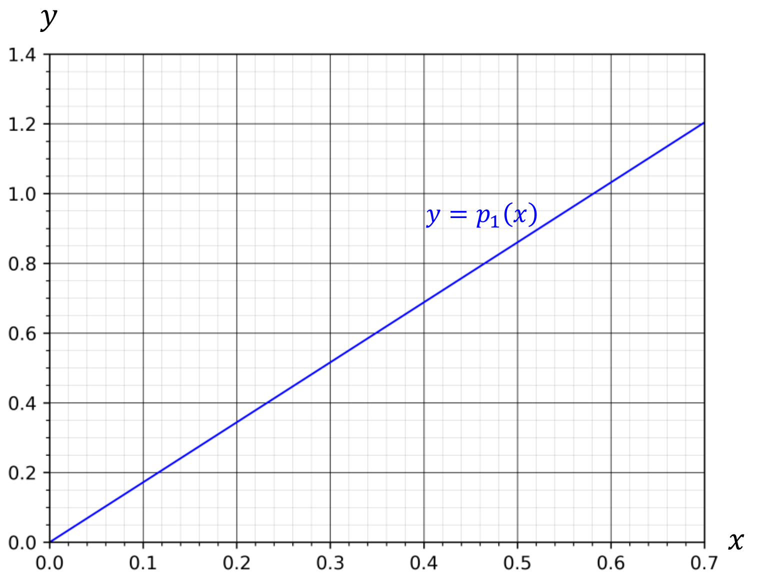
0	0.0	0.0000000
1	0.7	1.2039728

$$\begin{aligned} p_1(x) &= L_0^{(1)}(x) \cdot f(x_0) + L_1^{(1)}(x) \cdot f(x_1) \\ &= \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1) \end{aligned}$$

$$\begin{aligned} f(0.45) \approx p_1(0.45) &= \frac{0.45 - 0.7}{0.0 - 0.7} \cdot 0.0000000 + \frac{0.45 - 0.0}{0.7 - 0.0} \cdot 1.2039728 \\ &= 0.7739825 \end{aligned}$$

$$f(x) = -\ln(1 - x)$$

$$f(0.45) = -\ln(1 - 0.45) = 0.5978370$$



Lagrange Interpolating Polynomials

Example 04-02a: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

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1	0.7	1.2039728

$$p_1(x) = L_0^{(1)}(x) \cdot f(x_0) + L_1^{(1)}(x) \cdot f(x_1)$$
$$= \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1)$$
$$f(0.45) \approx p_1(0.45) = \frac{0.45 - 0.7}{0.0 - 0.7} \cdot 0.0000000 + \frac{0.45 - 0.0}{0.7 - 0.0} \cdot 1.2039728$$
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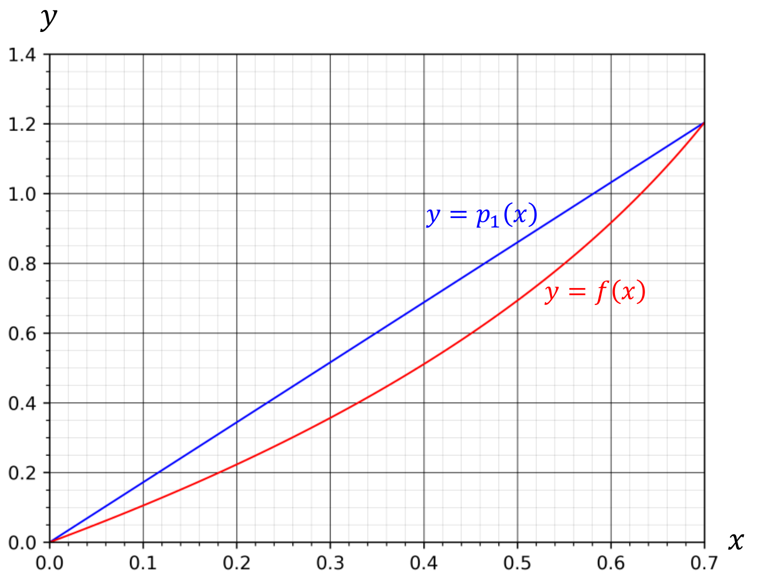

Lagrange Interpolating Polynomials

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$$p_1(x) = L_0^{(1)}(x) \cdot f(x_0) + L_1^{(1)}(x) \cdot f(x_1)$$
$$= \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1)$$

$$f(0.45) \approx p_1(0.45) = \frac{0.45 - 0.7}{0.0 - 0.7} \cdot 0.0000000 + \frac{0.45 - 0.0}{0.7 - 0.0} \cdot 1.2039728$$
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$$f(x) = -\ln(1 - x)$$
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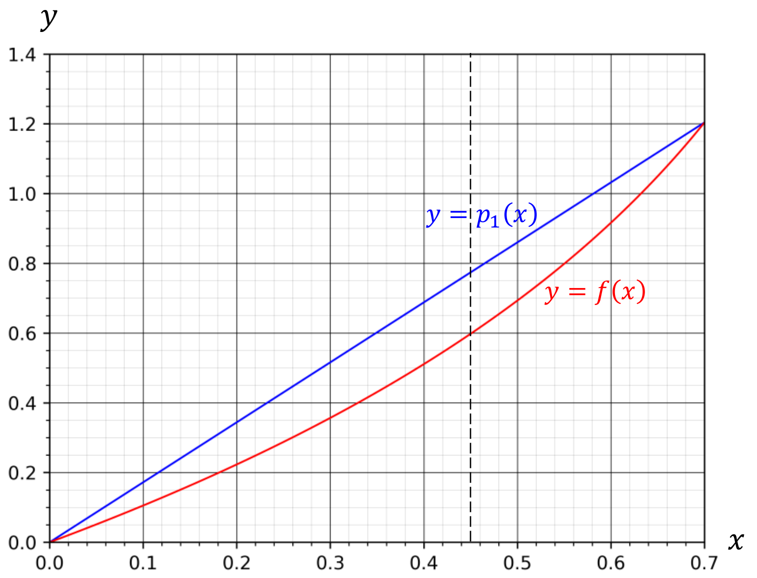

Lagrange Interpolating Polynomials

Example 04-02b: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749

Lagrange Interpolating Polynomials

Example 04-02b: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749

$$p_2(x) = L_0^{(2)}(x) \cdot f(x_0) + L_1^{(2)}(x) \cdot f(x_1) + L_2^{(2)}(x) \cdot f(x_2)$$

Lagrange Interpolating Polynomials

Example 04-02b: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749

$$p_2(x) = L_0^{(2)}(x) \cdot f(x_0) + L_1^{(2)}(x) \cdot f(x_1) + L_2^{(2)}(x) \cdot f(x_2)$$

$$= \frac{(x-x_1)\cdot(x-x_2)}{(x_0-x_1)\cdot(x_0-x_2)} \cdot f(x_0) + \frac{(x-x_0)\cdot(x-x_2)}{(x_1-x_0)\cdot(x_1-x_2)} \cdot f(x_1) + \frac{(x-x_0)\cdot(x-x_1)}{(x_2-x_0)\cdot(x_2-x_1)} \cdot f(x_2)$$

Lagrange Interpolating Polynomials

Example 04-02b: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
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$$p_2(x) = L_0^{(2)}(x)\cdot f(x_0) + L_1^{(2)}(x)\cdot f(x_1) + L_2^{(2)}(x)\cdot f(x_2)$$

$$= \frac{(x-x_1)\cdot(x-x_2)}{(x_0-x_1)\cdot(x_0-x_2)}\cdot f(x_0) + \frac{(x-x_0)\cdot(x-x_2)}{(x_1-x_0)\cdot(x_1-x_2)}\cdot f(x_1) + \frac{(x-x_0)\cdot(x-x_1)}{(x_2-x_0)\cdot(x_2-x_1)}\cdot f(x_2)$$

$$f(0.45) \approx p_2(0.45) = \frac{(0.45-0.7)\cdot(0.45-0.3)}{(0.0-0.7)\cdot(0.0-0.3)}\cdot 0.0000000 + \frac{(0.45-0.0)\cdot(0.45-0.3)}{(0.7-0.0)\cdot(0.7-0.3)}\cdot 1.2039728$$

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Lagrange Interpolating Polynomials

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$$f(0.45) \approx p_2(0.45) = \frac{(0.45-0.7)\cdot(0.45-0.3)}{(0.0-0.7)\cdot(0.0-0.3)}\cdot 0.0000000 + \frac{(0.45-0.0)\cdot(0.45-0.3)}{(0.7-0.0)\cdot(0.7-0.3)}\cdot 1.2039728$$

$$+\frac{(0.45-0.0)\cdot(0.45-0.7)}{(0.3-0.0)\cdot(0.3-0.7)}\cdot 0.3566749$$

$$= 0.6246262$$

Lagrange Interpolating Polynomials

Example 04-02b: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
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2	0.3	0.3566749

$$p_2(x) = L_0^{(2)}(x)\cdot f(x_0) + L_1^{(2)}(x)\cdot f(x_1) + L_2^{(2)}(x)\cdot f(x_2)$$

$$= \frac{(x-x_1)\cdot(x-x_2)}{(x_0-x_1)\cdot(x_0-x_2)}\cdot f(x_0) + \frac{(x-x_0)\cdot(x-x_2)}{(x_1-x_0)\cdot(x_1-x_2)}\cdot f(x_1) + \frac{(x-x_0)\cdot(x-x_1)}{(x_2-x_0)\cdot(x_2-x_1)}\cdot f(x_2)$$

$$f(0.45) \approx p_2(0.45) = \frac{(0.45-0.7)\cdot(0.45-0.3)}{(0.0-0.7)\cdot(0.0-0.3)}\cdot 0.0000000 + \frac{(0.45-0.0)\cdot(0.45-0.3)}{(0.7-0.0)\cdot(0.7-0.3)}\cdot 1.2039728$$

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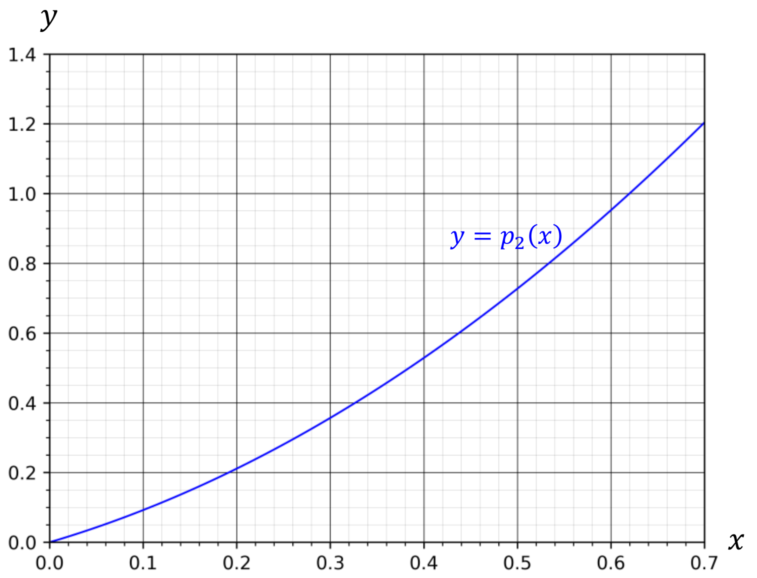
$$= 0.6246262$$



Lagrange Interpolating Polynomials

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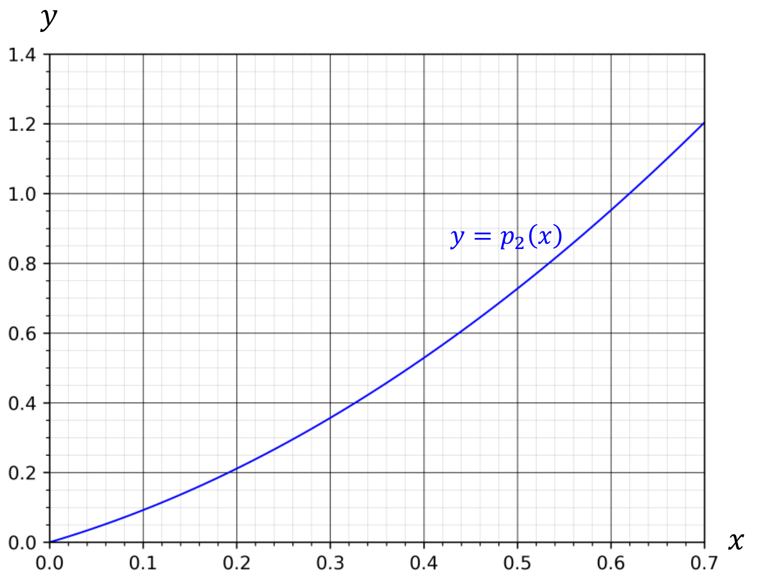

$$p_2(x) = L_0^{(2)}(x) \cdot f(x_0) + L_1^{(2)}(x) \cdot f(x_1) + L_2^{(2)}(x) \cdot f(x_2)$$

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$$f(0.45) \approx p_2(0.45) = \frac{(0.45-0.7)\cdot(0.45-0.3)}{(0.0-0.7)\cdot(0.0-0.3)} \cdot 0.0000000 + \frac{(0.45-0.0)\cdot(0.45-0.3)}{(0.7-0.0)\cdot(0.7-0.3)} \cdot 1.2039728$$

$$+ \frac{(0.45-0.0)\cdot(0.45-0.7)}{(0.3-0.0)\cdot(0.3-0.7)} \cdot 0.3566749$$

$$= 0.6246262$$

$$f(x) = -\ln(1-x)$$

$$f(0.45) = -\ln(1-0.45) = 0.5978370$$

Lagrange Interpolating Polynomials

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2	0.3	0.3566749

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$$= \frac{(x-x_1)\cdot(x-x_2)}{(x_0-x_1)\cdot(x_0-x_2)}\cdot f(x_0) + \frac{(x-x_0)\cdot(x-x_2)}{(x_1-x_0)\cdot(x_1-x_2)}\cdot f(x_1) + \frac{(x-x_0)\cdot(x-x_1)}{(x_2-x_0)\cdot(x_2-x_1)}\cdot f(x_2)$$

$$f(0.45) \approx p_2(0.45) = \frac{(0.45-0.7)\cdot(0.45-0.3)}{(0.0-0.7)\cdot(0.0-0.3)}\cdot 0.0000000 + \frac{(0.45-0.0)\cdot(0.45-0.3)}{(0.7-0.0)\cdot(0.7-0.3)}\cdot 1.2039728$$

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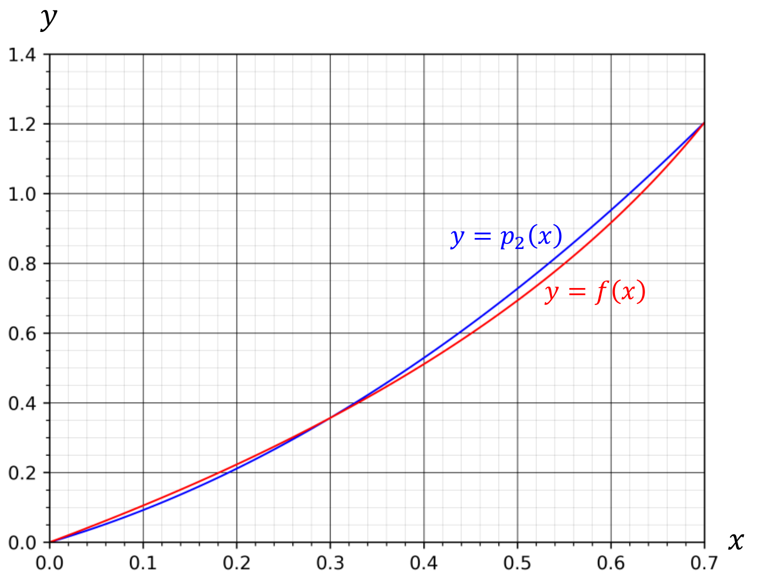

Lagrange Interpolating Polynomials

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$$p_2(x) = L_0^{(2)}(x) \cdot f(x_0) + L_1^{(2)}(x) \cdot f(x_1) + L_2^{(2)}(x) \cdot f(x_2)$$

$$= \frac{(x-x_1)\cdot(x-x_2)}{(x_0-x_1)\cdot(x_0-x_2)} \cdot f(x_0) + \frac{(x-x_0)\cdot(x-x_2)}{(x_1-x_0)\cdot(x_1-x_2)} \cdot f(x_1) + \frac{(x-x_0)\cdot(x-x_1)}{(x_2-x_0)\cdot(x_2-x_1)} \cdot f(x_2)$$

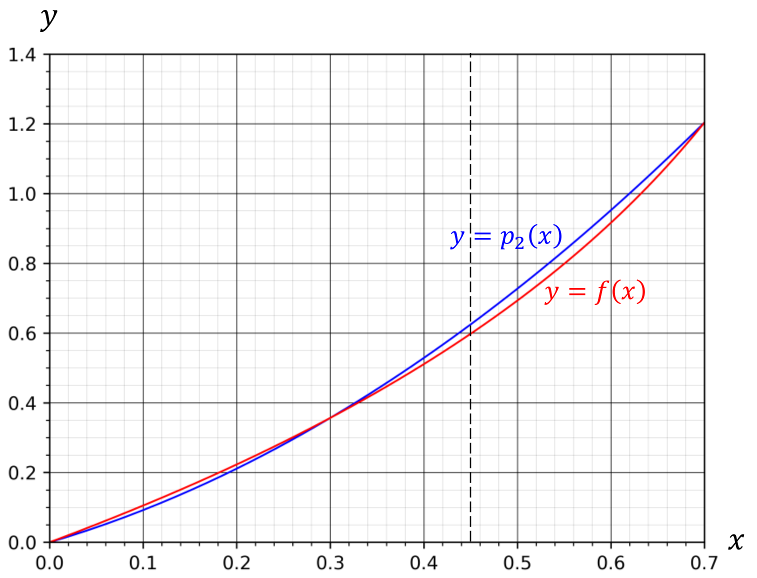
$$f(0.45) \approx p_2(0.45) = \frac{(0.45-0.7)\cdot(0.45-0.3)}{(0.0-0.7)\cdot(0.0-0.3)} \cdot 0.0000000 + \frac{(0.45-0.0)\cdot(0.45-0.3)}{(0.7-0.0)\cdot(0.7-0.3)} \cdot 1.2039728$$

$$+ \frac{(0.45-0.0)\cdot(0.45-0.7)}{(0.3-0.0)\cdot(0.3-0.7)} \cdot 0.3566749$$

$$= 0.6246262$$

$$f(x) = -\ln(1-x)$$

$$f(0.45) = -\ln(1-0.45) = 0.5978370$$



Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

$$p_3(x) = L_0^{(3)}(x) \cdot f(x_0) + L_1^{(3)}(x) \cdot f(x_1) + L_2^{(3)}(x) \cdot f(x_2) + L_3^{(3)}(x) \cdot f(x_3)$$

Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

$$p_3(x) = L_0^{(3)}(x) \cdot f(x_0) + L_1^{(3)}(x) \cdot f(x_1) + L_2^{(3)}(x) \cdot f(x_2) + L_3^{(3)}(x) \cdot f(x_3)$$

$$\begin{aligned} f(0.45) \approx p_3(0.45) = & \frac{(0.45-0.7)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.0-0.7)\cdot(0.0-0.3)\cdot(0.0-0.2)} \cdot 0.0000000 \\ & + \frac{(0.45-0.0)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.7-0.0)\cdot(0.7-0.3)\cdot(0.7-0.2)} \cdot 1.2039728 \\ & + \frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.2)}{(0.3-0.0)\cdot(0.3-0.7)\cdot(0.3-0.2)} \cdot 0.3566749 \\ & + \frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.3)}{(0.2-0.0)\cdot(0.2-0.7)\cdot(0.2-0.3)} \cdot 0.2231436 \end{aligned}$$

Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

$$p_3(x) = L_0^{(3)}(x)\cdot f(x_0) + L_1^{(3)}(x)\cdot f(x_1) + L_2^{(3)}(x)\cdot f(x_2) + L_3^{(3)}(x)\cdot f(x_3)$$

$$\begin{aligned}
f(0.45) \approx p_3(0.45) &= \frac{(0.45-0.7)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.0-0.7)\cdot(0.0-0.3)\cdot(0.0-0.2)}\cdot 0.0000000 \\
&+\frac{(0.45-0.0)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.7-0.0)\cdot(0.7-0.3)\cdot(0.7-0.2)}\cdot 1.2039728 \\
&+\frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.2)}{(0.3-0.0)\cdot(0.3-0.7)\cdot(0.3-0.2)}\cdot 0.3566749 \\
&+\frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.3)}{(0.2-0.0)\cdot(0.2-0.7)\cdot(0.2-0.3)}\cdot 0.2231436 \\
&= 0.6045237
\end{aligned}$$

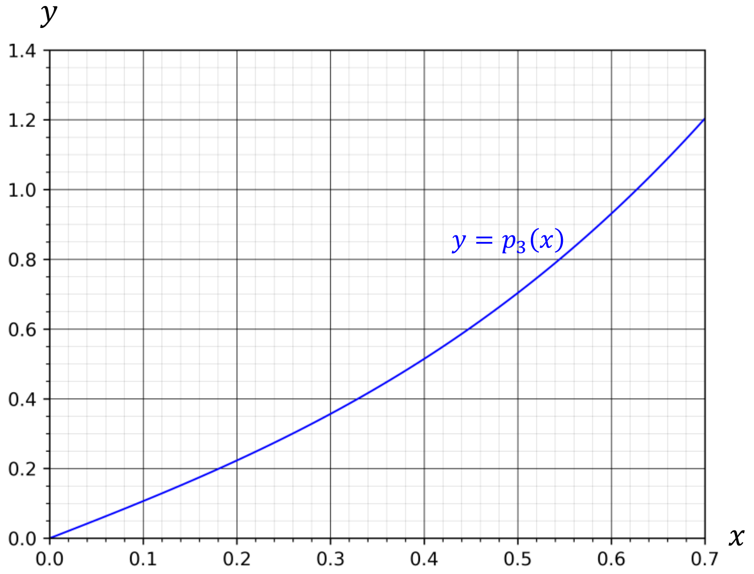
Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

$$p_3(x) = L_0^{(3)}(x)\cdot f(x_0) + L_1^{(3)}(x)\cdot f(x_1) + L_2^{(3)}(x)\cdot f(x_2) + L_3^{(3)}(x)\cdot f(x_3)$$

$$\begin{aligned} f(0.45) \approx p_3(0.45) &= \frac{(0.45-0.7)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.0-0.7)\cdot(0.0-0.3)\cdot(0.0-0.2)}\cdot 0.0000000 \\ &+\frac{(0.45-0.0)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.7-0.0)\cdot(0.7-0.3)\cdot(0.7-0.2)}\cdot 1.2039728 \\ &+\frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.2)}{(0.3-0.0)\cdot(0.3-0.7)\cdot(0.3-0.2)}\cdot 0.3566749 \\ &+\frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.3)}{(0.2-0.0)\cdot(0.2-0.7)\cdot(0.2-0.3)}\cdot 0.2231436 \\ &= 0.6045237 \end{aligned}$$



Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
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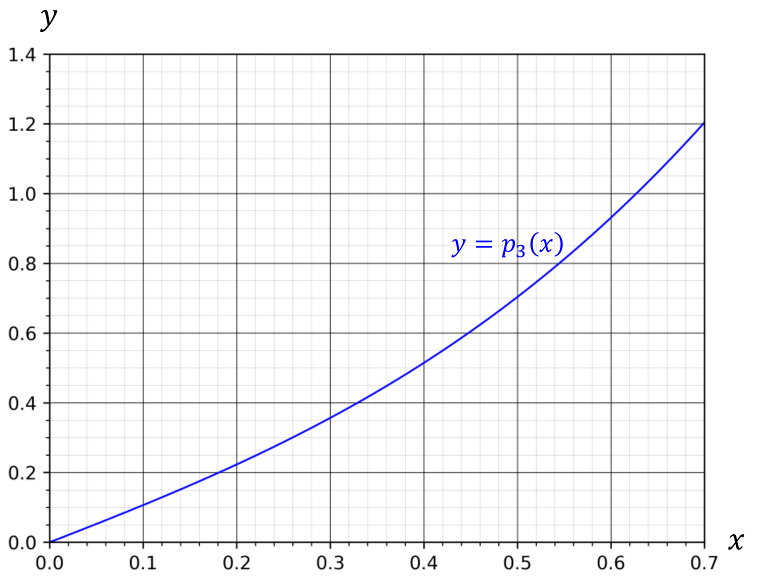
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436



$$p_3(x) = L_0^{(3)}(x) \cdot f(x_0) + L_1^{(3)}(x) \cdot f(x_1) + L_2^{(3)}(x) \cdot f(x_2) + L_3^{(3)}(x) \cdot f(x_3)$$

$$\begin{aligned} f(0.45) \approx p_3(0.45) &= \frac{(0.45-0.7)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.0-0.7)\cdot(0.0-0.3)\cdot(0.0-0.2)} \cdot 0.0000000 \\ &+ \frac{(0.45-0.0)\cdot(0.45-0.3)\cdot(0.45-0.2)}{(0.7-0.0)\cdot(0.7-0.3)\cdot(0.7-0.2)} \cdot 1.2039728 \\ &+ \frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.2)}{(0.3-0.0)\cdot(0.3-0.7)\cdot(0.3-0.2)} \cdot 0.3566749 \\ &+ \frac{(0.45-0.0)\cdot(0.45-0.7)\cdot(0.45-0.3)}{(0.2-0.0)\cdot(0.2-0.7)\cdot(0.2-0.3)} \cdot 0.2231436 \\ &= 0.6045237 \end{aligned}$$

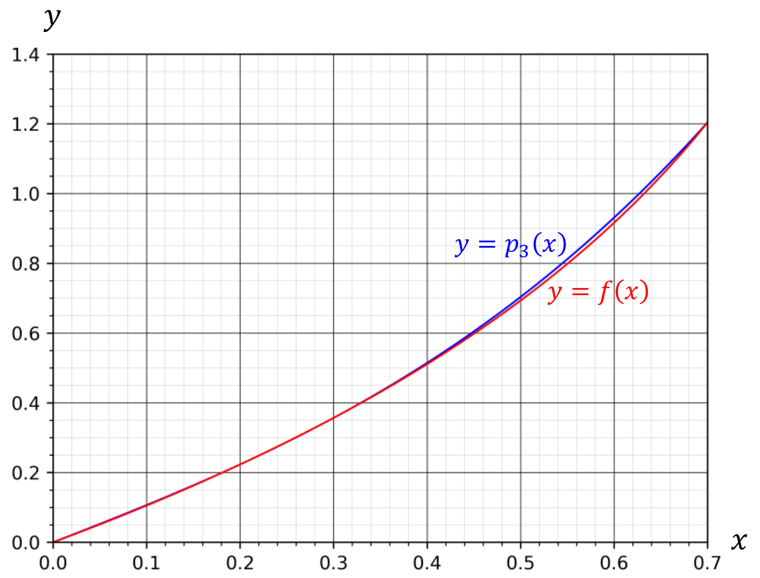
$$f(x) = -\ln(1-x)$$

$$f(0.45) = -\ln(1-0.45) = 0.5978370$$

Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436



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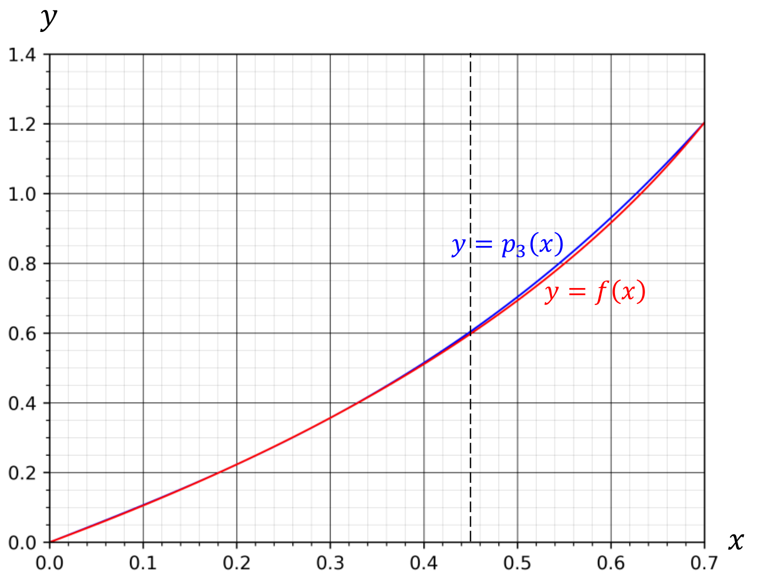

$$p_3(x) = L_0^{(3)}(x)\cdot f(x_0) + L_1^{(3)}(x)\cdot f(x_1) + L_2^{(3)}(x)\cdot f(x_2) + L_3^{(3)}(x)\cdot f(x_3)$$

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